Compiler Construction

Chapter 0

Course Organisation

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Lectures and Labs

Time slot 1:

- Tuesday 14–18: CZS3 SR 130
- Lecture: 1 hour (wall clock time)
- Break: 15 minutes
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- Break: 15 minutes
- Project lab: 1 hour (wall clock time)
- ... starting soon

Time slot 2:

- Thursday 16–18: CIP SR 1086
- Lab / Übung
- ... soon (stay tuned)

Formalities

Numbers:

- 9 EC / LP ("Leistungspunkte")
- 6 SWS ("Semesterwochenstunden")
- **New in 2025 !!**
- ... considered successful
- ... to be continued in 2026

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Evaluation:

- 60% CiviC compiler project
- 30% Assignments (weekly to bi-weekly)
- 10% CiviC compiler report

Course Contents

Chapters:

1. Introduction and Motivation
2. Compilers at a glance: bird's eye view tour through a compiler
3. Lexical analysis: from characters to words
4. Syntactic analysis: from words to sentences and intermediate representation
5. Semantic analysis: consistency and type inference / checking
6. Optimisation: make code better: time, energy, size, ...
7. Code generation: the final step

Compiler Project

CiviC: "Civilised C":

- Model language (toy language?)
- C-like syntax
- Basic types, usual operators
- Loops, if-then-else, ...
- Functions

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Language core plus extensions:

- Extension 1: local function definitions
- Extension 2: arrays (1-dimensional vectors)
- Extension 3: multi-dimensional arrays

CiviC Compiler Project

CiviC tool chain:

- Reference compiler(s)
- CiviC Assembler
- CiviC Virtual Machine (simplified JVM)
- ... novel implementations in Rust as of 2026 (thanks to Eric)

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First steps:

- Learn how to program in CiviC
- Introduction to CiviC next week in lecture
- Get CiviC tool chain installed
- First assignment for labs next week

CoCoNut Compiler Framework

CoCoNut: "Compiler Construction in a Nutshell":

- Originally developed in my group at University of Amsterdam (UvA)
- 10+ Bachelor projects
- Integrated by Damian Fröhlich (UvA)
- Now continued at FSU by Eric Wintzler, Arnold Krause ...

CoCoNut Compiler Framework

What is CoCoNut ?

- Based on C language !!
- **If your C skills are not top-notch, you have about 2 weeks time ...**
- Tailor-made DSL (domain-specific language) describes:
  - data structures
  - compiler passes (sequence of steps)
  - where in data structure to do something ("the where")
  - build system (based on Cmake)
- CoCoNut compiler generates (a lot of) C code
- You fill in the concrete gaps ("the what") in C
- Introduction to CoCoNut in 2 weeks time

Literature

Mandatory:

- Nothing, really.
- Just join the lectures and take notes
- Read the slides for reference
- And, no, I haven't written a compiler textbook myself

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Optional:

- Of course, there are great books on compilers
- ... more next week
- Good news: they are quite stable
- ... unlike much other stuff in computer science or artificial intelligence

The End: Questions ?

